



5. Rowhammer

Materials from Onur Mutlu, ETH Zurich

Jin Hong

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How Reliable/Secure/Safe is This Bridge?



How Secure Are These People?



How Safe & Secure Are Our Platforms?



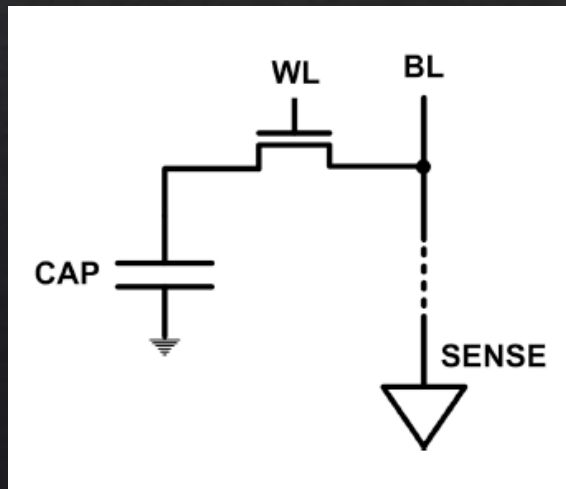
What Is RowHammer?

- ◇ One can predictably induce bit flips in commodity DRAM chips
 - ◇ >80% of the tested DRAM chips are vulnerable
- ◇ First example of how a simple hardware failure mechanism can create a widespread system security vulnerability



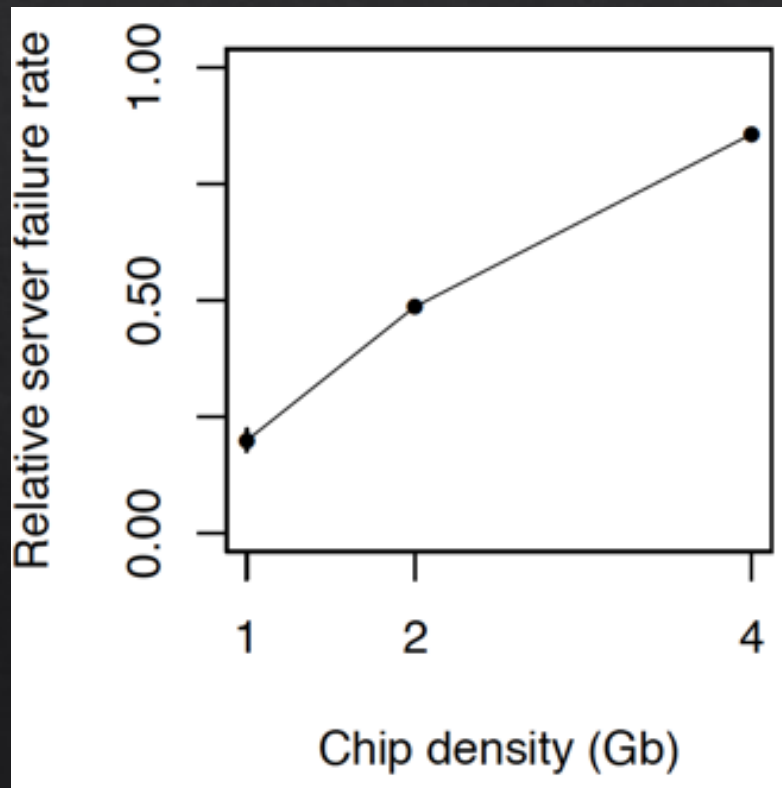
The DRAM Scaling Problem

- ◇ DRAM stores charge in a capacitor (charge-based memory)
 - ◇ Capacitor must be large enough for reliable sensing
 - ◇ Access transistor should be large enough for low leakage and high retention time
 - ◇ Scaling beyond 40-35nm (2013) is challenging [ITRS, 2009]
- ◇ DRAM capacity, cost, and energy/power hard to scale

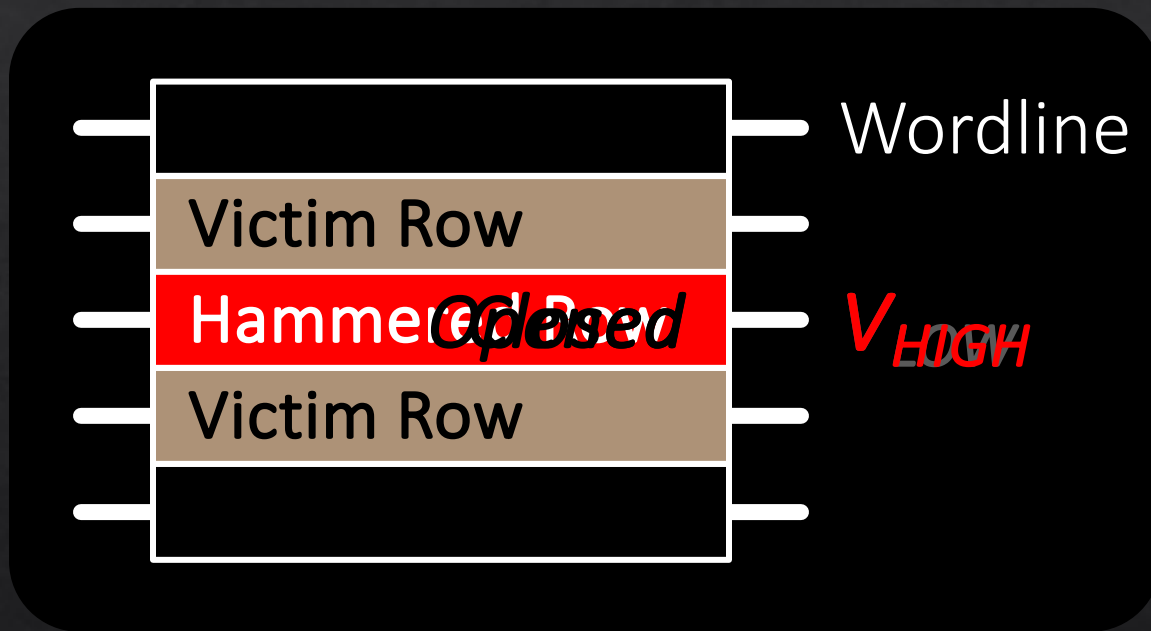


As Memory Scales, It Becomes Unreliable

- ◇ Data from all of Facebook's servers worldwide
 - ◇ Meza+, "Revisiting Memory Errors in Large-Scale Production Data Centers," DSN'15.



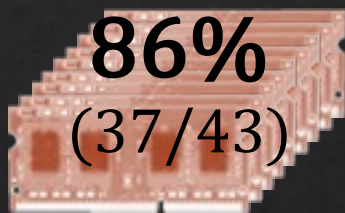
Modern DRAM is Prone to Disturbance Errors



Repeatedly reading a row enough times (before memory gets refreshed) induces **disturbance errors** in **adjacent rows** in **most real DRAM chips you can buy today**

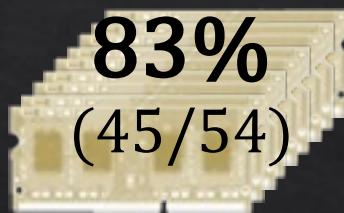
Most DRAM Modules Are Vulnerable

A company



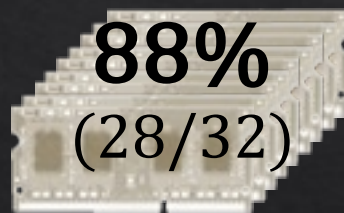
Up to
 1.0×10^7
errors

B company



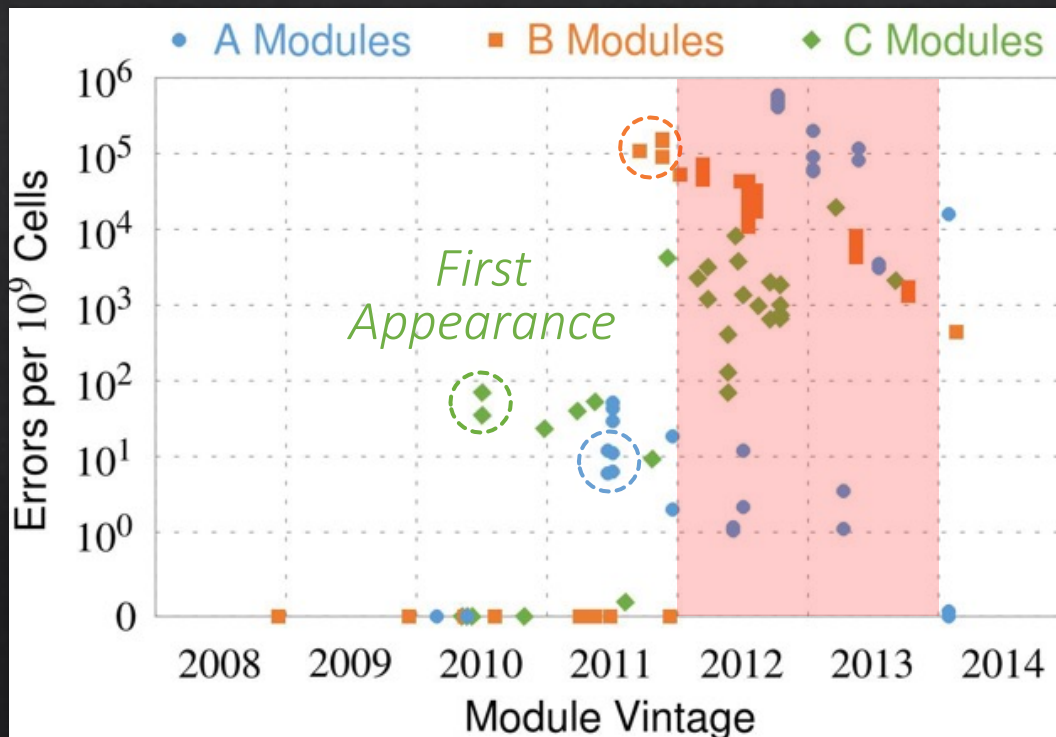
Up to
 2.7×10^6
errors

C company



Up to
 3.3×10^5
errors

Recent DRAM Is More Vulnerable

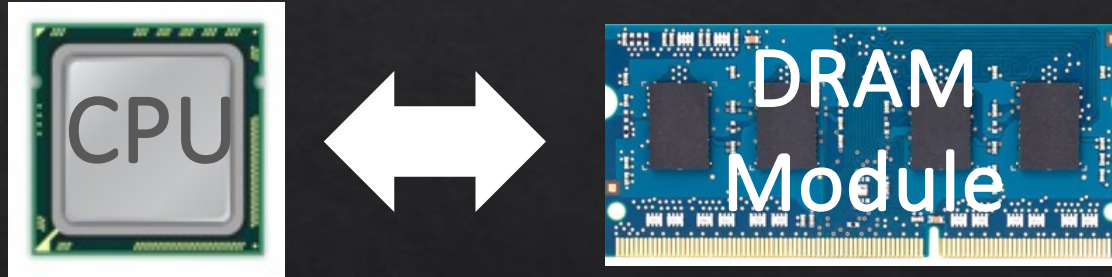


All modules from 2012-2013 are vulnerable

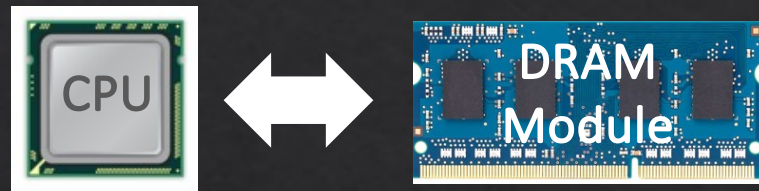
Why Is This Happening?

- ◇ DRAM cells are too close to each other!
 - ◇ They are not electrically isolated from each other
- ◇ Access to one cell affects the value in nearby cells
 - ◇ due to [electrical interference](#) between
 - ◇ the cells
 - ◇ wires used for accessing the cells
 - ◇ Also called cell-to-cell coupling/interference
- ◇ Example: When we activate (apply high voltage) to a row, an adjacent row gets slightly activated as well
 - ◇ Vulnerable cells in that slightly-activated row lose a little bit of charge
 - ◇ If RowHammer happens enough times, charge in such cells gets drained

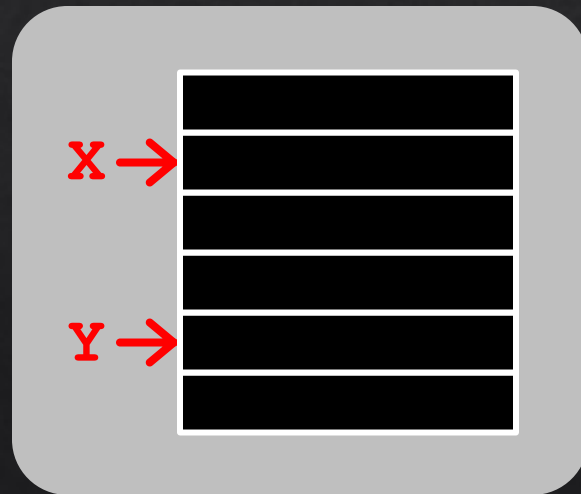
A Simple Program Can Induce Many Errors



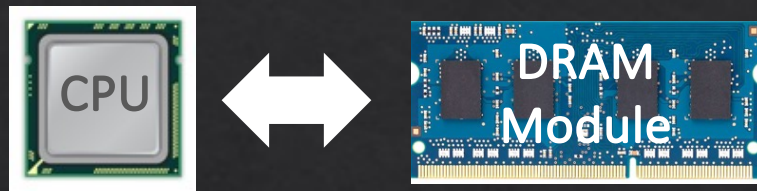
A Simple Program Can Induce Many Errors



```
loop:  
  mov  (X), %eax  
  mov  (Y), %ebx  
  clflush (X)  
  clflush (Y)  
  mfence  
  jmp  loop
```

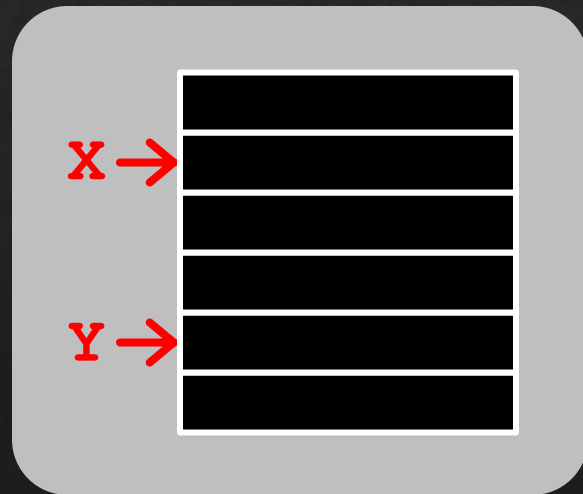


A Simple Program Can Induce Many Errors

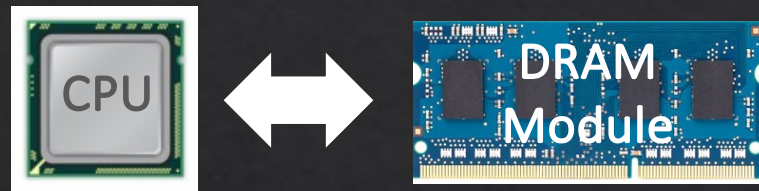


1. Avoid *cache hits*
 - Flush **X** from cache

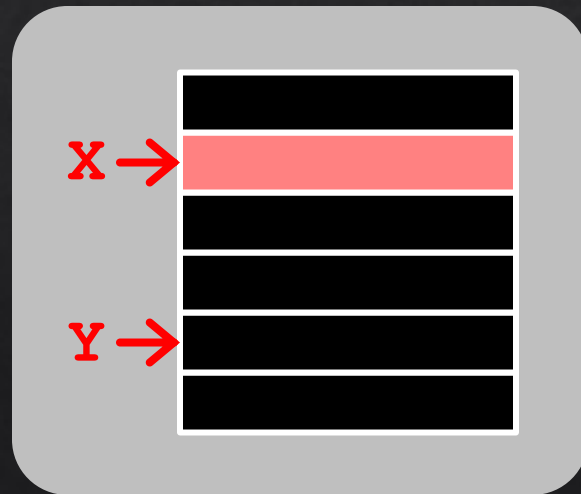
2. Avoid *row hits* to **X**
 - Read **Y** in another row



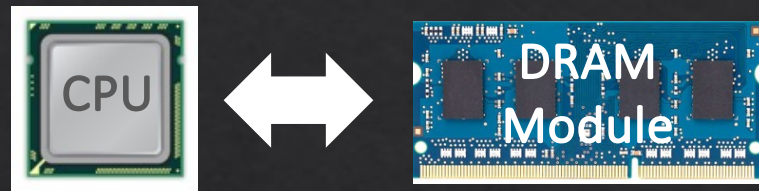
A Simple Program Can Induce Many Errors



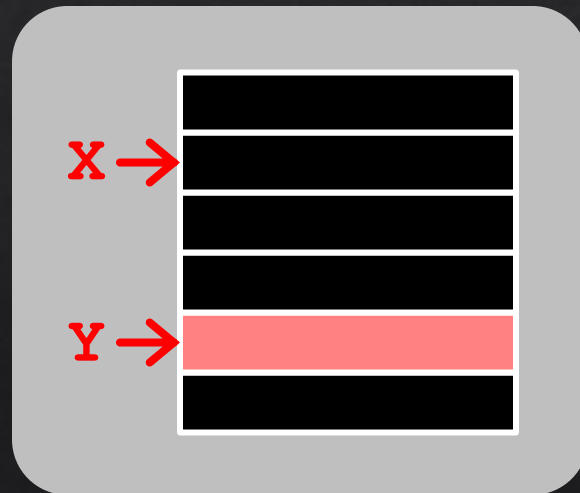
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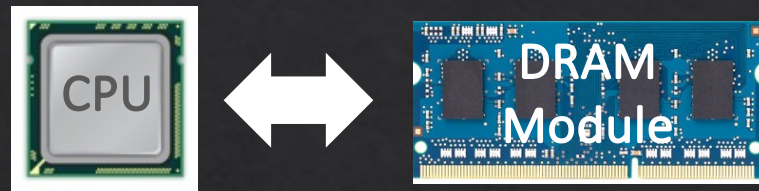
A Simple Program Can Induce Many Errors



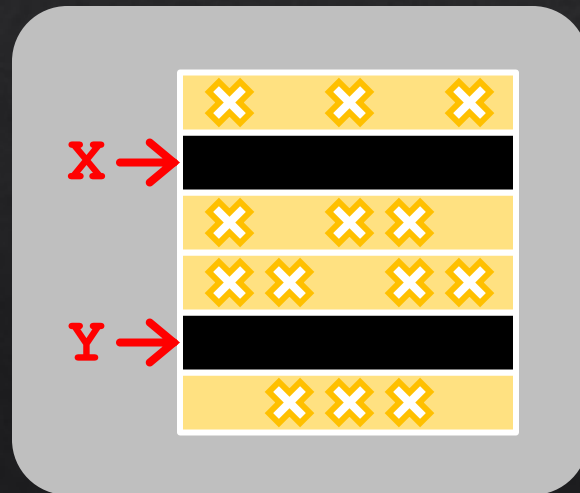
```
loop:  
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  mov (Y), %ebx  
  clflush (X)  
  clflush (Y)  
  mfence  
  jmp loop
```



A Simple Program Can Induce Many Errors



```
loop:  
  mov (X), %eax  
  mov (Y), %ebx  
  clflush (X)  
  clflush (Y)  
  mfence  
  jmp loop
```



Observed Errors in Real Systems

CPU Architecture	Errors	Access-Rate
Intel Haswell (2013)	22.9K	12.3M/sec
Intel Ivy Bridge (2012)	20.7K	11.7M/sec
Intel Sandy Bridge (2011)	16.1K	11.6M/sec
AMD Piledriver (2012)	59	6.1M/sec

Security Implications



More Security Implications (I)

"We can gain unrestricted access to systems of website visitors."

Not there yet, but ...



ROOT privileges for web apps!

29

Daniel Gruss (@lavados), Clémentine Maurice (@BloodyTangerine),
December 28, 2015 — 32c3, Hamburg, Germany

www.laik.tugraz.at



GATED
COMMUNITIES

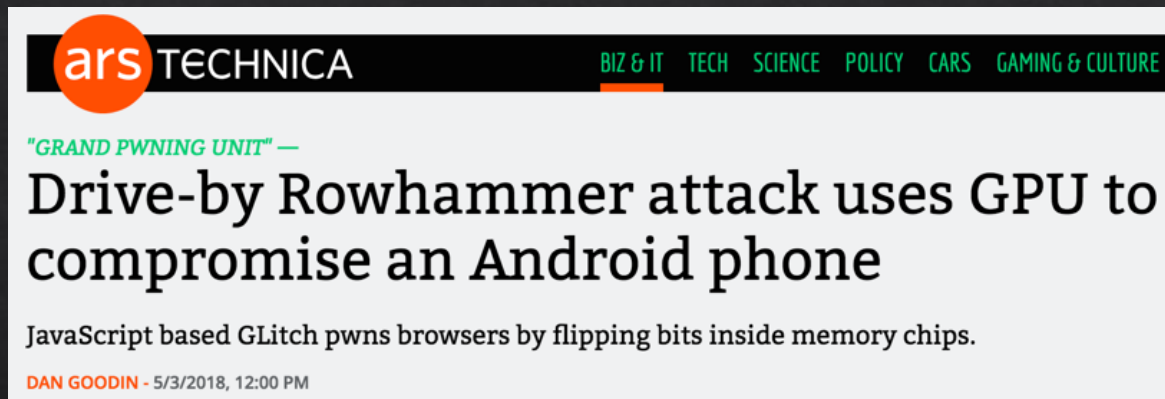
Rowhammer.js: A Remote Software-Induced Fault Attack in JavaScript (DIMVA'16)

More Security Implications (II)



More Security Implications (III)

- ◇ Using an integrated GPU in a mobile system to remotely escalate privilege via the WebGL interface - IEEE S&P 2018



ars TECHNICA

BIZ & IT TECH SCIENCE POLICY CARS GAMING & CULTURE

"GRAND PWINING UNIT" —

Drive-by Rowhammer attack uses GPU to compromise an Android phone

JavaScript based GLitch pwns browsers by flipping bits inside memory chips.

DAN GOODIN - 5/3/2018, 12:00 PM

Grand Pwning Unit: Accelerating Microarchitectural Attacks with the GPU

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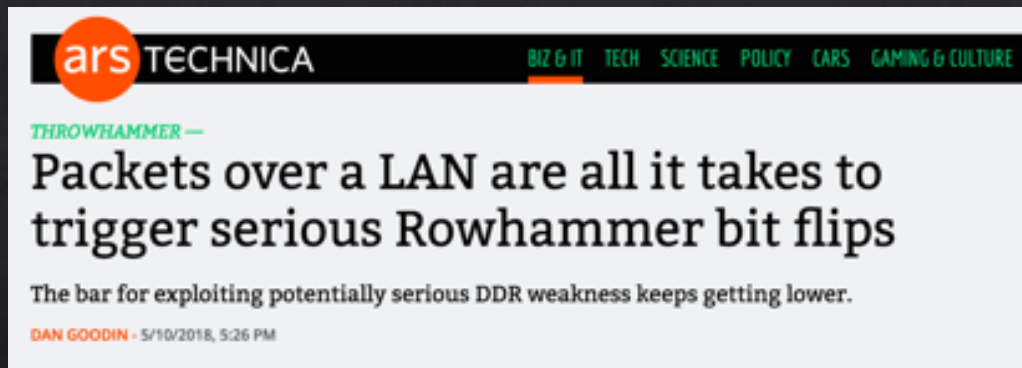
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More Security Implications (IV)

◇ Rowhammer over RDMA (I) - USENIX ATC 2018



Throwhammer: Rowhammer Attacks over the Network and Defenses

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University of Cyprus

Cristiano Giuffrida
VU Amsterdam

Herbert Bos
VU Amsterdam

Kaveh Razavi
VU Amsterdam

More Security Implications (V)

◇ Rowhammer over RDMA (II)



Nethammer: Inducing Rowhammer Faults through Network Requests

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Graz University of Technology

Daniel Gruss
Graz University of Technology

Misiker Tadesse Aga
University of Michigan

Clémentine Maurice
Univ Rennes, CNRS, IRISA

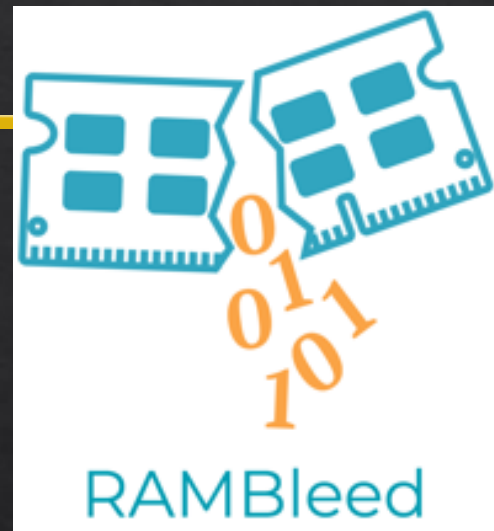
Lukas Lamster
Graz University of Technology

Michael Schwarz
Graz University of Technology

Lukas Raab
Graz University of Technology

More Security Implications (VI)

◇ IEEE S&P 2020



RAMBleed: Reading Bits in Memory Without Accessing Them

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More Security Implications (VII)

◇ USENIX Security 2019



A Single Bit-flip Can Cause Terminal Brain Damage to DNNs

One specific bit-flip in a DNN's representation leads to accuracy drop over 90%

Our research found that a specific bit-flip in a DNN's bitwise representation can cause the accuracy loss up to 90%, and the DNN has 40-50% parameters, on average, that can lead to the accuracy drop over 10% when individually subjected to such single bitwise corruptions...

[Read More](#)

Terminal Brain Damage: Exposing the Graceless Degradation in Deep Neural Networks Under Hardware Fault Attacks

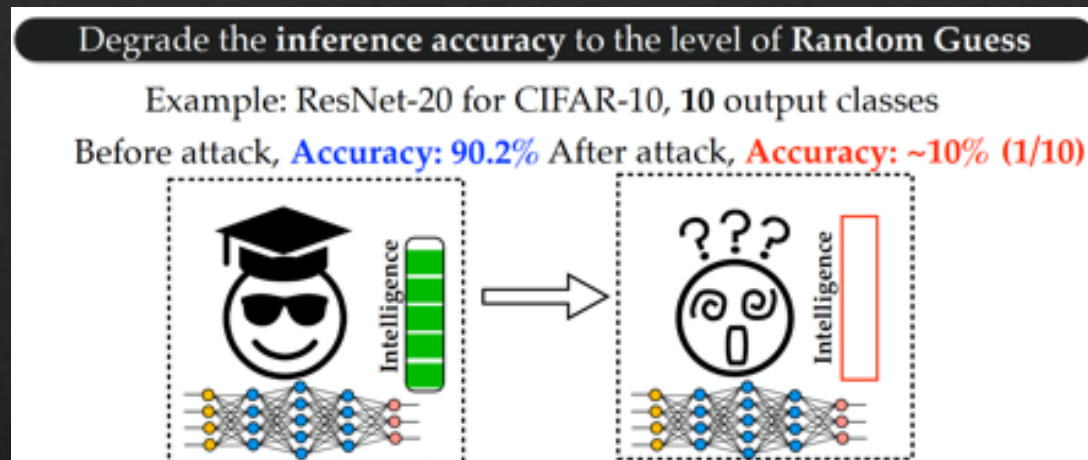
Sanghyun Hong, Pietro Frigo[†], Yiğitcan Kaya, Cristiano Giuffrida[†], Tudor Dumitraş

University of Maryland, College Park

[†]*Vrije Universiteit Amsterdam*

More Security Implications (VIII)

◇ USENIX Security 2020



DeepHammer: Depleting the Intelligence of Deep Neural Networks through Targeted Chain of Bit Flips

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Arizona State University
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More Security Implications (IX)

- ◇ Rowhammer on MLC NAND Flash (based on [Cai+, HPCA 2017])



Security

Rowhammer RAM attack adapted to hit flash storage

Project Zero's two-year-old dog learns a new trick

By Richard Chirgwin 17 Aug 2017 at 04:27

17 SHARE ▼

From random block corruption to privilege escalation: A filesystem attack vector for rowhammer-like attacks

Anil Kurmus

Nikolas Ioannou

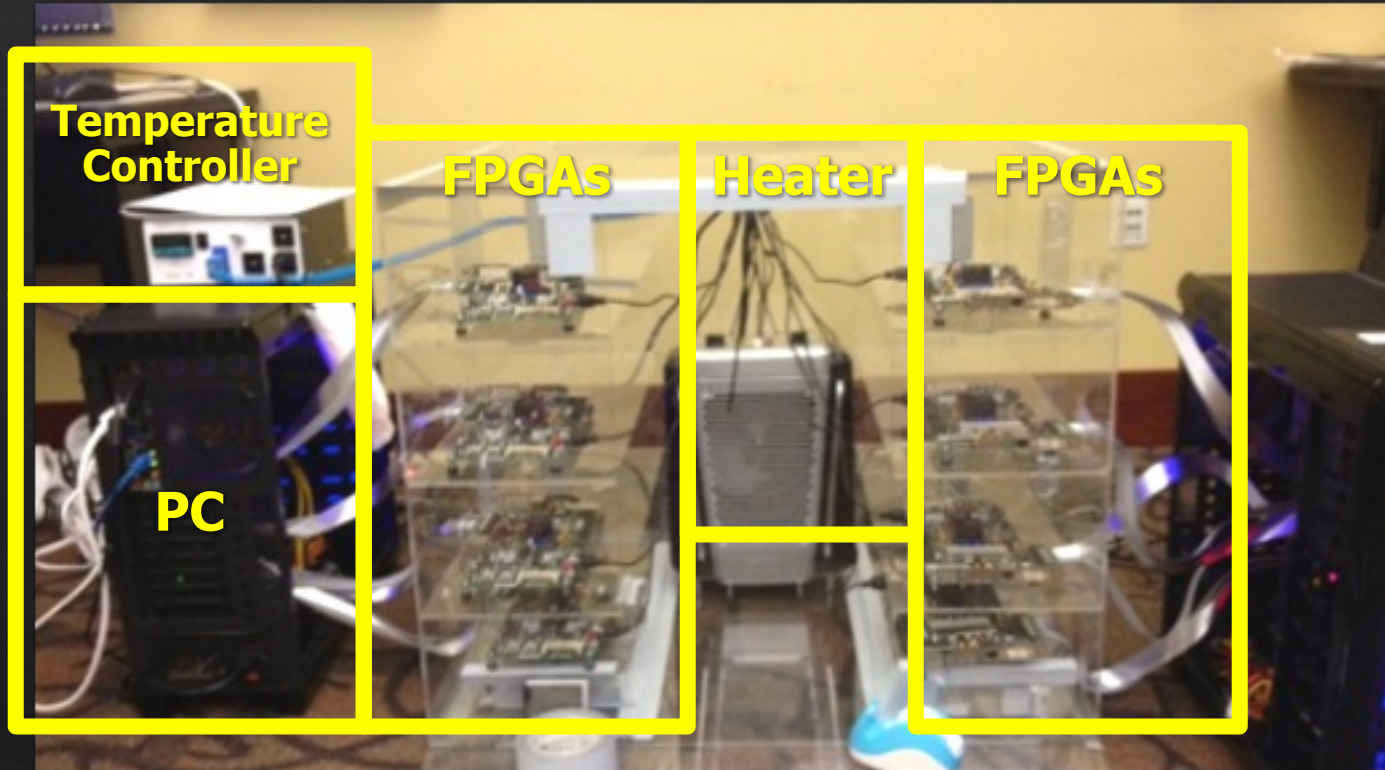
Matthias Neugschwandtner

Nikolaos Papandreou

Thomas Parnell

IBM Research – Zurich

RowHammer Experiment



Tested DRAM Modules from 2008-2014 (129 total)

Manufacturer	Module	Date*	Timing [†]		Organization		Chip			Victims-per-Module				R _{1,0k} (ms)
		(yy-mm)	Freq (MT/s)	t _{AC} (ns)	Size (GB)	Chips	Size (Gb/s) [‡]	Pin	DieVersion [§]	Average	Minimum	Maximum	Min	
A	Total of 43 Modules	A ₁	10-08	1066	50.625	0.5	4	1	×16	B	0	0	0	–
		A ₂	10-20	1066	50.625	1	8	1	×8	F	0	0	0	–
		A _{2.5}	10-20	1066	50.625	0.5	4	1	×16	B	0	0	0	–
		A _{6.3}	11-24	1066	49.125	1	4	2	×16	D	7.8 × 10 ³	5.2 × 10 ³	1.0 × 10 ²	21.3
		A _{8.12}	11-26	1066	49.125	1	4	2	×16	D	2.4 × 10 ²	5.4 × 10 ³	4.4 × 10 ²	16.4
		A _{12.14}	11-50	1066	49.125	1	4	2	×16	D	8.8 × 10 ³	1.7 × 10 ³	1.6 × 10 ²	26.2
		A _{16.16}	12-22	1600	50.625	1	4	2	×16	D	9.5	9	1.0 × 10 ³	34.4
		A _{17.18}	12-26	1600	49.125	2	8	2	×8	M	1.2 × 10 ²	3.7 × 10 ³	2.0 × 10 ²	21.3
		A _{19.30}	12-40	1600	48.125	2	8	2	×8	K	8.6 × 10 ⁶	7.0 × 10 ⁶	1.0 × 10 ⁷	8.2
		A _{21.34}	13-02	1600	48.125	2	8	2	×8	–	1.8 × 10 ⁶	1.0 × 10 ⁶	3.5 × 10 ⁶	11.5
		A _{26.36}	13-14	1600	48.125	2	8	2	×8	–	4.0 × 10 ³	1.9 × 10 ³	6.1 × 10 ³	21.3
		A _{27.38}	13-20	1600	48.125	2	8	2	×8	K	1.7 × 10 ⁶	1.4 × 10 ⁶	2.0 × 10 ⁶	9.8
		A _{39.40}	13-28	1600	48.125	2	8	2	×8	K	5.7 × 10 ⁴	5.4 × 10 ⁴	6.0 × 10 ⁴	16.4
		A ₄₁	14-04	1600	49.125	2	8	2	×8	–	2.7 × 10 ³	2.7 × 10 ³	2.7 × 10 ³	18.0
A _{42.43}	14-04	1600	48.125	2	8	2	×8	K	0.5	0	1	62.3		
B	Total of 54 Modules	B ₁	08-49	1066	50.625	1	8	1	×8	D	0	0	0	–
		B ₂	09-49	1066	50.625	1	8	1	×8	E	0	0	0	–
		B ₃	10-19	1066	50.625	1	8	1	×8	F	0	0	0	–
		B ₄	10-31	1333	49.125	2	8	2	×8	C	0	0	0	–
		B ₅	11-13	1333	49.125	2	8	2	×8	C	0	0	0	–
		B ₆	11-16	1066	50.625	1	8	1	×8	F	0	0	0	–
		B ₇	11-19	1066	50.625	1	8	1	×8	F	0	0	0	–
		B ₈	11-25	1333	49.125	2	8	2	×8	C	0	0	0	–
		B ₉	11-37	1333	49.125	2	8	2	×8	D	1.9 × 10 ⁶	1.9 × 10 ⁶	1.9 × 10 ⁶	11.5
		B _{10.12}	11-46	1333	49.125	2	8	2	×8	D	2.2 × 10 ⁶	1.5 × 10 ⁶	2.7 × 10 ⁶	11.5
		B ₁₃	11-49	1333	49.125	2	8	2	×8	C	0	0	0	–
		B ₁₄	12-01	1866	47.125	2	8	2	×8	D	9.1 × 10 ³	9.1 × 10 ³	9.1 × 10 ³	9.8
		B _{15.31}	12-10	1866	47.125	2	8	2	×8	D	9.8 × 10 ³	7.8 × 10 ³	1.2 × 10 ⁶	11.5
		B _{16.31}	12-25	1600	48.125	2	8	2	×8	E	7.4 × 10 ³	7.4 × 10 ³	7.4 × 10 ³	11.5
		B _{17.32}	12-28	1600	48.125	2	8	2	×8	E	5.2 × 10 ³	1.9 × 10 ³	7.3 × 10 ³	11.5
		B _{18.42}	12-31	1600	48.125	2	8	2	×8	E	4.0 × 10 ³	2.9 × 10 ³	5.5 × 10 ³	13.1
C	Total of 32 Modules	B _{19.47}	13-19	1600	48.125	2	8	2	×8	E	1.1 × 10 ³	7.4 × 10 ⁴	1.4 × 10 ³	14.7
		B _{22.53}	13-40	1333	49.125	2	8	2	×8	D	2.6 × 10 ⁴	2.3 × 10 ⁴	2.9 × 10 ⁴	21.3
		B ₂₄	14-07	1333	49.125	2	8	2	×8	D	7.5 × 10 ³	7.5 × 10 ³	7.5 × 10 ³	26.2
		C ₁	10-18	1333	49.125	2	8	2	×8	A	0	0	0	–
		C ₂	10-20	1066	50.625	2	8	2	×8	A	0	0	0	–
		C ₃	10-22	1066	50.625	2	8	2	×8	A	0	0	0	–
		C _{4.5}	10-26	1333	49.125	2	8	2	×8	B	8.9 × 10 ²	6.0 × 10 ²	1.2 × 10 ³	29.5
		C ₆	10-43	1333	49.125	1	8	1	×8	T	0	0	0	–
		C ₇	10-51	1333	49.125	2	8	2	×8	B	4.0 × 10 ²	4.0 × 10 ²	4.0 × 10 ²	29.5
		C ₈	11-12	1333	46.25	2	8	2	×8	B	6.9 × 10 ²	6.9 × 10 ²	6.9 × 10 ²	21.3
		C ₉	11-19	1333	46.25	2	8	2	×8	B	9.2 × 10 ²	9.2 × 10 ²	9.2 × 10 ²	27.9
		C ₁₀	11-31	1333	49.125	2	8	2	×8	B	3	3	3	39.3
		C ₁₁	11-42	1333	49.125	2	8	2	×8	B	1.6 × 10 ²	1.6 × 10 ²	1.6 × 10 ²	39.3
		C ₁₂	11-48	1600	48.125	2	8	2	×8	C	7.1 × 10 ⁴	7.1 × 10 ⁴	7.1 × 10 ⁴	19.7
		C ₁₃	12-08	1333	49.125	2	8	2	×8	C	3.9 × 10 ⁴	3.9 × 10 ⁴	3.9 × 10 ⁴	21.3
		C _{14.15}	12-12	1333	49.125	2	8	2	×8	C	3.7 × 10 ⁴	2.1 × 10 ⁴	5.4 × 10 ⁴	21.3

* We report the manufacture date marked on the chip packages, which is more accurate than other dates that can be gleaned from a module.

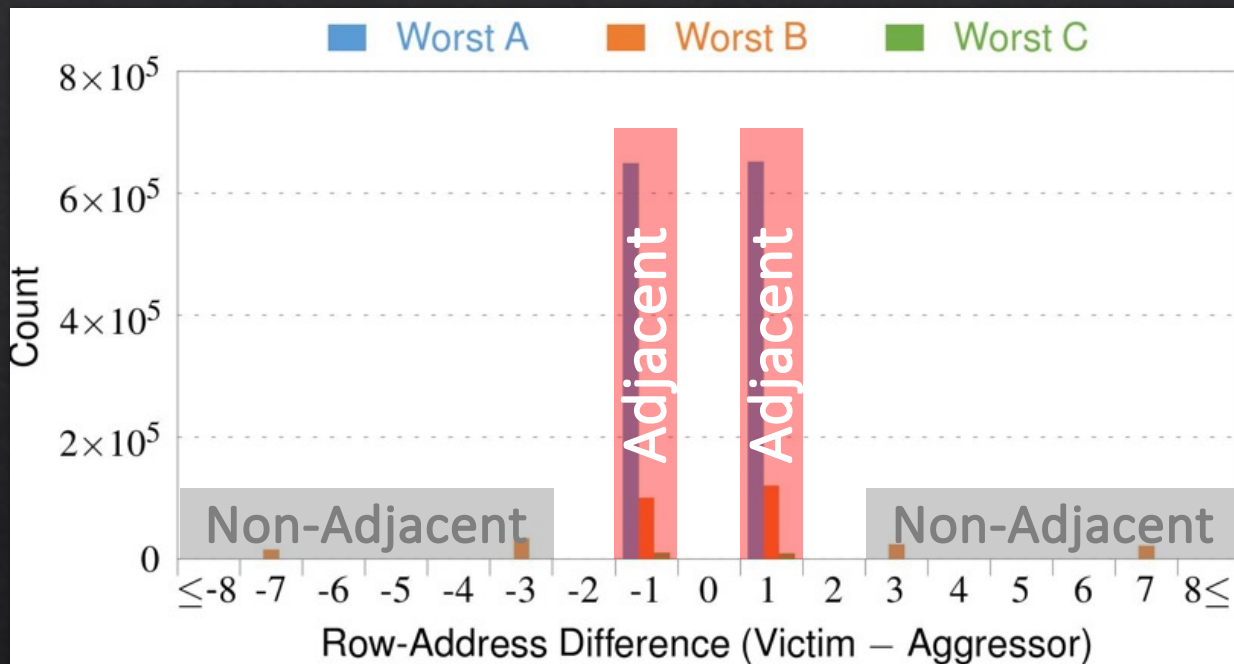
† We report timing constraints stored in the module's on-board ROM [33], which is read by the system BIOS to calibrate the memory controller.

‡ The maximum DRAM chip size supported by our testing platform is 2Gb.

§ We report DRAM die versions marked on the chip packages, which typically progress in the following manner: A → A → B → C → ...

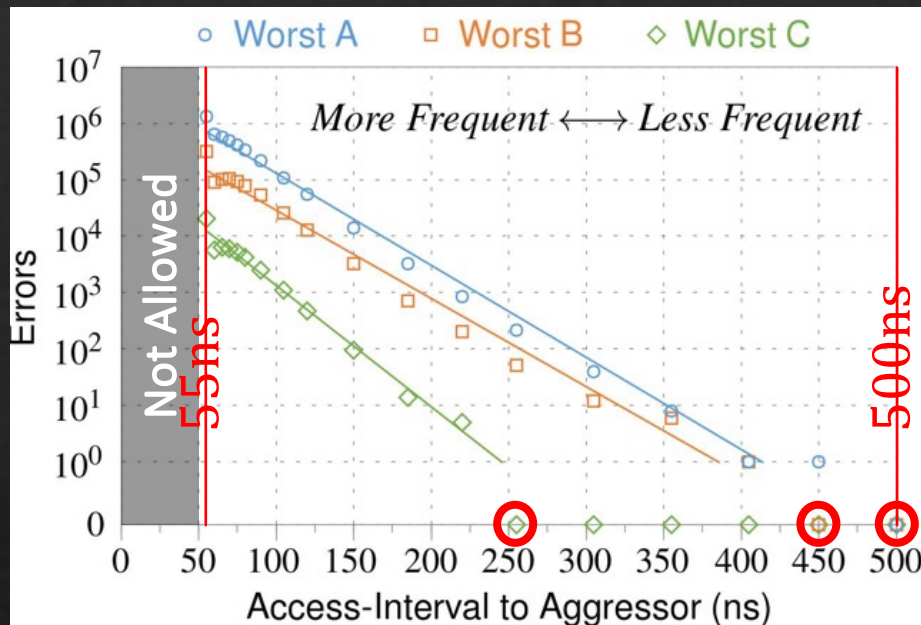
Table 3. Sample population of 129 DDR3 DRAM modules, categorized by manufacturer and sorted by manufacture date

Adjacency: Aggressor & Victim



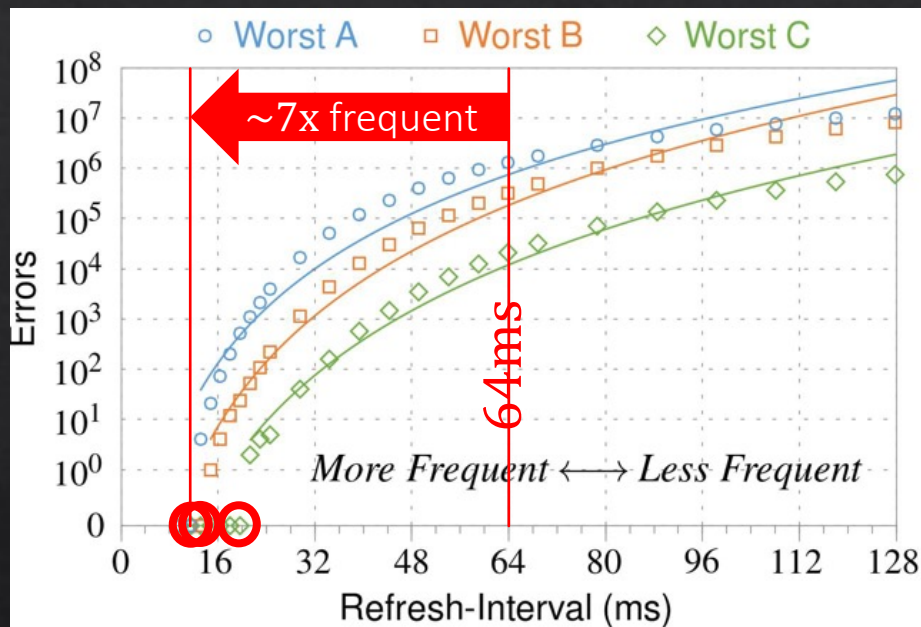
Note: Using three modules with the most errors (only first bank)

1 Access Interval (Aggressor)



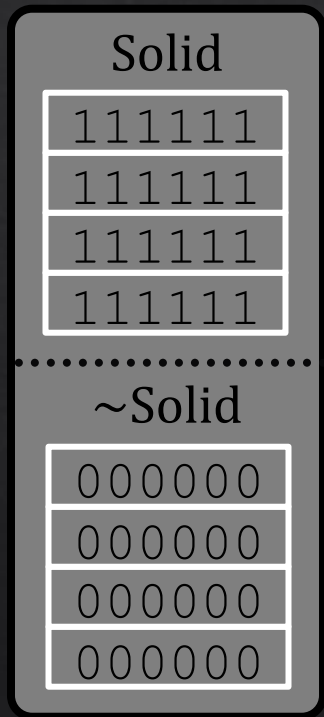
Note: Using three modules with the most errors (only first bank)

2 Refresh Interval



Note: Using three modules with the most errors (only first bank)

3 Data Pattern



Other Key Observations

- ◇ *Victim Cells $\neq$ Retention-Weak Cells*
 - ◇ Almost no overlap between them
- ◇ *Errors are repeatable*
 - ◇ Across ten iterations of testing, >70% of victim cells had errors in every iteration
- ◇ *As many as 4 errors per cache-line*
 - ◇ Simple ECC (e.g., SECDED) cannot prevent all errors
- ◇ *Cells affected by two aggressors on either side*
 - ◇ Double sided hammering

RowHammer is Getting Much Worse

Revisiting RowHammer: An Experimental Analysis of Modern DRAM Devices and Mitigation Techniques

Jeremie S. Kim^{§†} Minesh Patel[§] A. Giray Yağlıkçı[§]
Hasan Hassan[§] Roknoddin Azizi[§] Lois Orosa[§] Onur Mutlu^{§†}
[§]*ETH Zürich* [†]*Carnegie Mellon University*

(2020)

Understanding RowHammer Under Reduced Wordline Voltage: An Experimental Study Using Real DRAM Devices

A. Giray Yağlıkçı¹ Haocong Luo¹ Geraldo F. de Oliveira¹ Ataberk Olgun¹ Minesh Patel¹
Jisung Park¹ Hasan Hassan¹ Jeremie S. Kim¹ Lois Orosa^{1,2} Onur Mutlu¹
¹*ETH Zürich* ²*Galicia Supercomputing Center (CESGA)*

(2022)

An Experimental Analysis of RowHammer in HBM2 DRAM Chips

Ataberk Olgun¹ Majd Osseiran^{1,2} A. Giray Yağlıkçı¹ Yahya Can Tuğrul¹
Haocong Luo¹ Steve Rhyner¹ Behzad Salami¹ Juan Gomez Luna¹ Onur Mutlu¹
¹*SAFARI Research Group, ETH Zürich* ²*American University of Beirut*

(2023)

RowHammer Solutions

Naive Solutions

1 *Throttle accesses to same row*

- ◇ Limit access-interval: $\geq 500\text{ns}$
- ◇ Limit number of accesses: $\leq 128\text{K}$ ($= 64\text{ms}/500\text{ns}$)

2 *Refresh more frequently*

- ◇ Shorten refresh-interval by $\sim 7\text{x}$

Two Types of RowHammer Solutions

- ◆ Immediate
 - ◆ To protect the vulnerable DRAM chips in the field
 - ◆ Limited possibilities
- ◆ Longer-term
 - ◆ To protect future DRAM chips
 - ◆ Wider range of protection mechanisms

Some Potential Solutions (ISCA 2014)

- Make better DRAM chips

Cost

- Refresh frequently

Power, Performance

- Sophisticated ECC

Cost, Power

- Access counters

Cost, Power, Complexity

Apple's Security Patch for RowHammer

◇ <https://support.apple.com/en-gb/HT204934>

Available for: OS X Mountain Lion v10.8.5, OS X Mavericks v10.9.5

Impact: A malicious application may induce memory corruption to escalate privileges

Description: A disturbance error, also known as Rowhammer, exists with some DDR3 RAM that could have led to memory corruption. This issue was mitigated by increasing memory refresh rates.

CVE-ID

CVE-2015-3693 : Mark Seaborn and Thomas Dullien of Google, working from original research by Yoongu Kim et al (2014)

HP, Lenovo, and many other vendors released similar patches

RowHammer Solution: PARA

- ◇ *PARA: Probabilistic Adjacent Row Activation*
- ◇ Key Idea
 - ◇ After closing a row, we activate (i.e., refresh) one of its neighbors with a low probability: $p = 0.005$
- ◇ Reliability Guarantee
 - ◇ When $p=0.005$, errors in one year: 9.4×10^{-14}
 - ◇ By adjusting the value of p , we can vary the strength of protection against errors

Advantages of PARA

- ◇ *PARA refreshes rows infrequently*

- ◇ Low power

- ◇ Low performance-overhead

- ◇ Average slowdown: 0.20% (for 29 benchmarks)

- ◇ Maximum slowdown: 0.75%

- ◇ *PARA is stateless*

- ◇ Low cost

- ◇ Low complexity

- ◇ *PARA is an effective and low-overhead solution to prevent disturbance errors*

What next?

RowPress vs. RowHammer

Instead of using a high activation count,

👉 increase the time that the aggressor row stays open

RowHammer
Aggressor Row



RowPress
Aggressor Row



We observe bitflips even with **ONLY ONE activation** in extreme cases where the row stays open for 30ms

Architecting Future Memory for Security

- ◆ **Understand:** Methods for vulnerability modeling & discovery
 - ◆ Modeling and prediction based on real (device) data and analysis
 - ◆ Understanding vulnerabilities
 - ◆ Developing reliable metrics
- ◆ **Architect:** Principled architectures with security as key concern
 - ◆ Good partitioning of duties across the stack
 - ◆ Cannot give up performance and efficiency
 - ◆ Patch-ability in the field
- ◆ **Design & Test:** Principled design, automation, (online) testing
 - ◆ Design for security
 - ◆ High coverage and good interaction with system reliability methods

Two Major Future RowHammer Directions

◆ Understanding RowHammer

- ◆ Many effects still need to be rigorously examined
 - ◆ Aging of DRAM Chips
 - ◆ Environmental Conditions
 - ◆ Memory Access Patterns
 - ◆ Memory Controller & System Design Decisions
 - ◆ ...

◆ Solving RowHammer

- ◆ Flexible and Efficient RowHammer Solutions are necessary
 - ◆ In-field patchable / reconfigurable / programmable solutions
- ◆ Co-architecting System and Memory is important
 - ◆ To avoid performance and denial-of-service problems

Google's Original RowHammer Attack

Presented by Mark Seaborn and Thomas Dullien at BlackHat 2015

RowHammer Attack

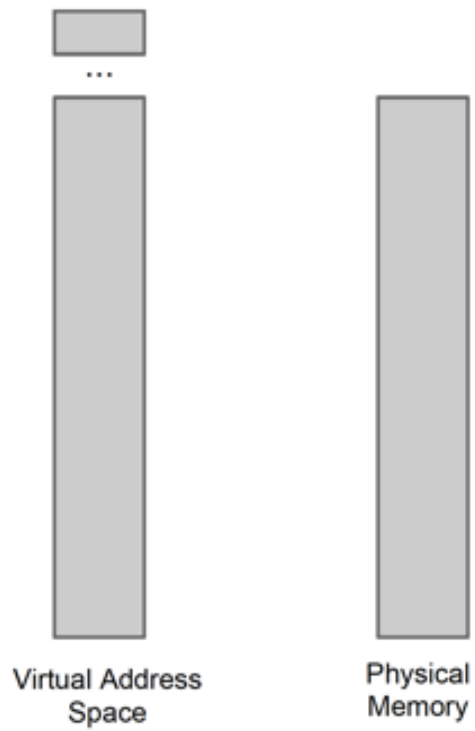
Kernel exploit

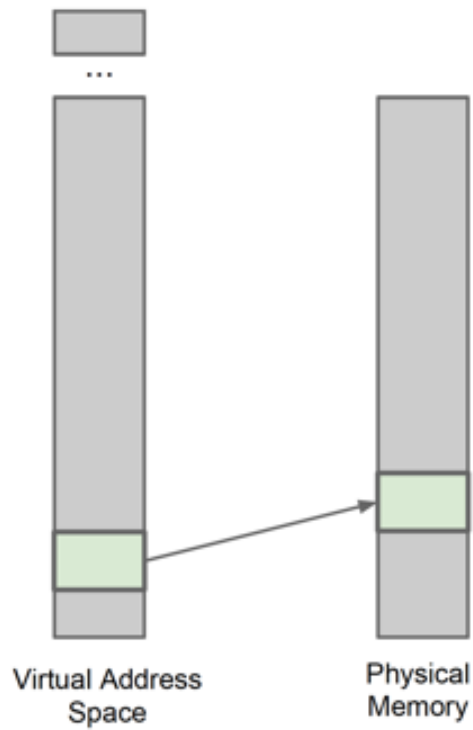
- x86 page tables entries (PTEs) are **dense and trusted**
 - They control access to physical memory
 - A bit flip in a PTE's physical page number can give a process access to a different physical page
- Aim of exploit: Get access to a page table
 - Gives access to all of physical memory
- Maximise chances that a bit flip is useful:
 - Spray physical memory with page tables
 - Check for useful, repeatable bit flip first

x86-64 Page Table Entries (PTEs)

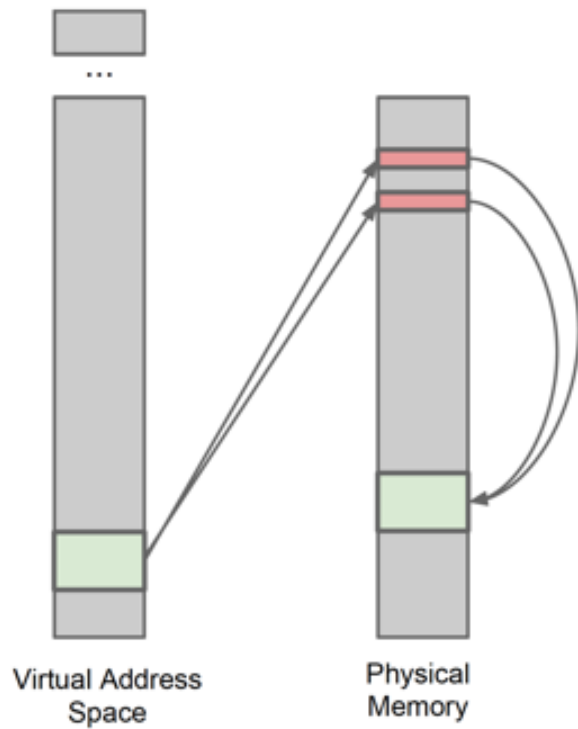
- | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|----------------------------|-----------|-------|--|----|--|--|--|--|--|--|--|-----|--|---|-----|-------------|-----|---|-----|---|-----|-------------|-----|-------------|--|-------------|--|-------------|--|---|--|
| 63 62 | | 52 51 | | 32 | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| N | Available | | Physical-Page Base Address
(This is an architectural limit. A given implementation may support fewer bits.) | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| X | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 31 | | | 12 11 | | | | | | | | | | | | 9 8 | | 7 6 | | 5 4 | | 3 2 | | 1 0 | | | | | | | | |
| Physical-Page Base Address | | | | | | | | | | | | AVL | | G | | P
A
T | | D | | A | | P
C
D | | P
W
T | | U
/
S | | R
/
W | | P | |

- Could flip:
 - "Writable" permission bit (RW): 1 bit → 2% chance
 - Physical page number: 20 bits on 4GB system → 31% chance

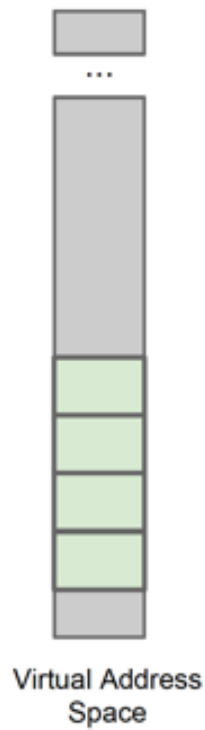




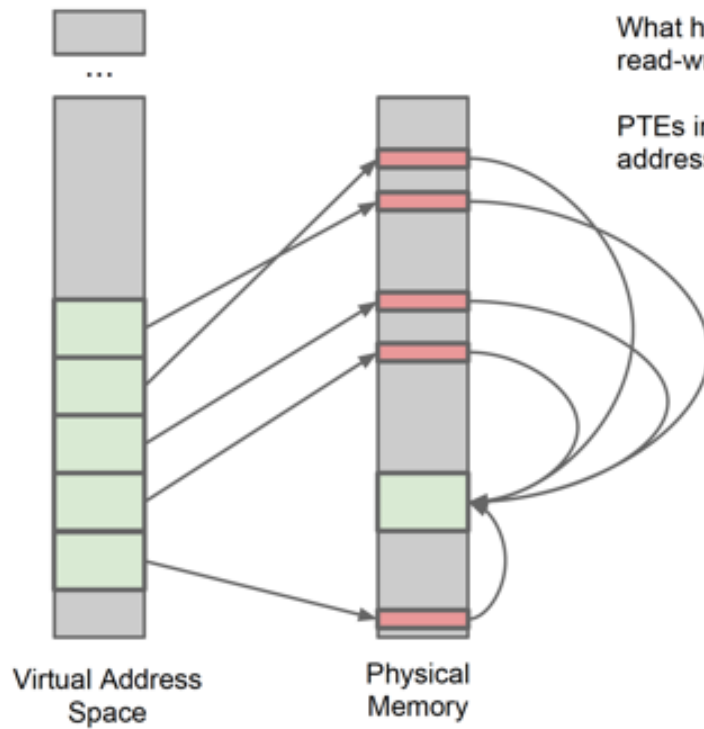
What happens when we map a file with read-write permissions?



What happens when we map a file with read-write permissions? Indirection via page tables.

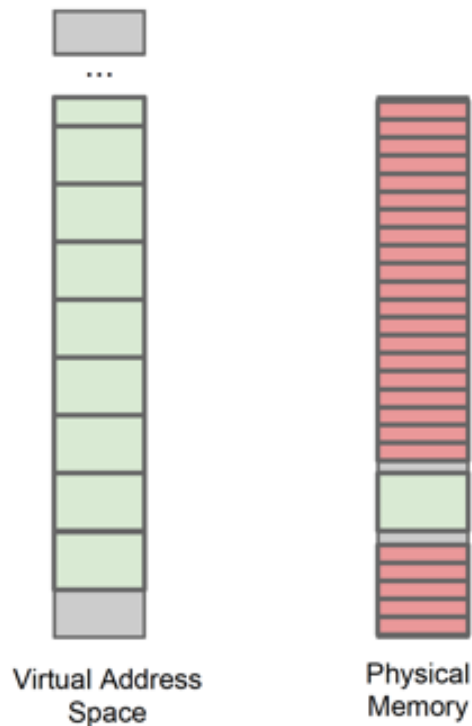


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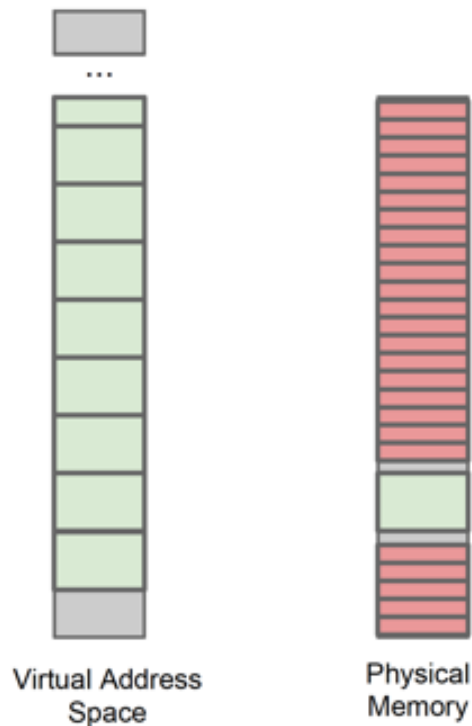
PTEs in physical memory help resolve virtual addresses to physical pages.



What happens when we repeatedly map a file with read-write permissions?

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We can fill physical memory with PTEs.

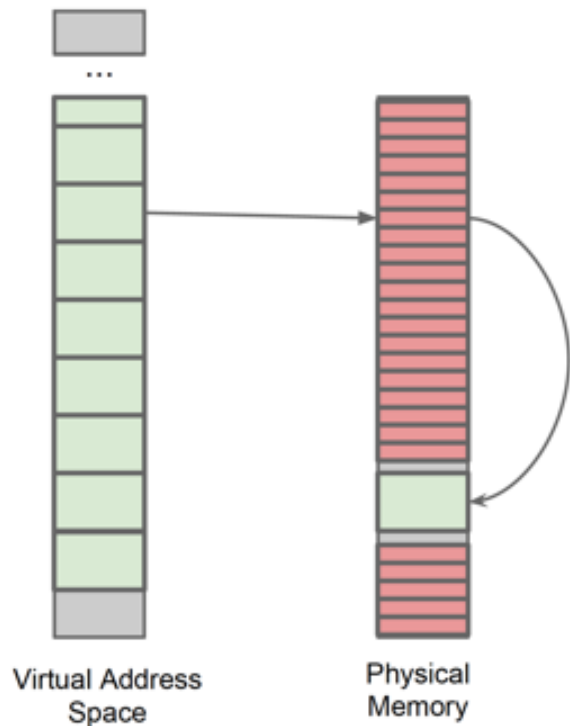


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We can fill physical memory with PTEs.

Each of them points to pages in the same physical file mapping.



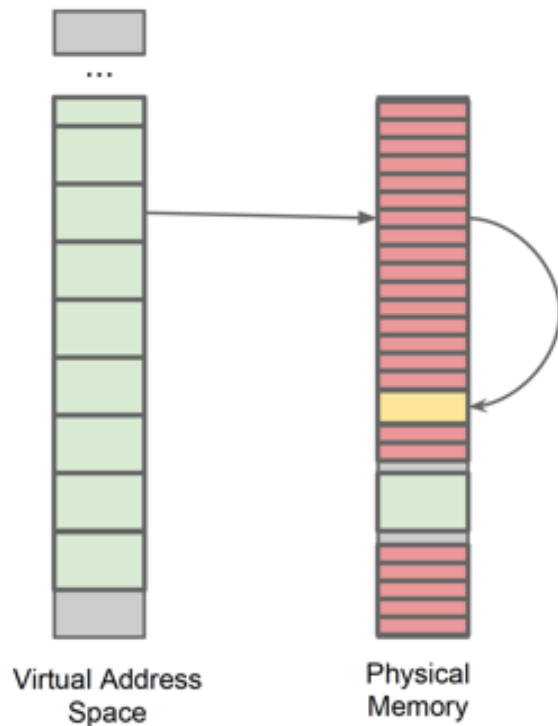
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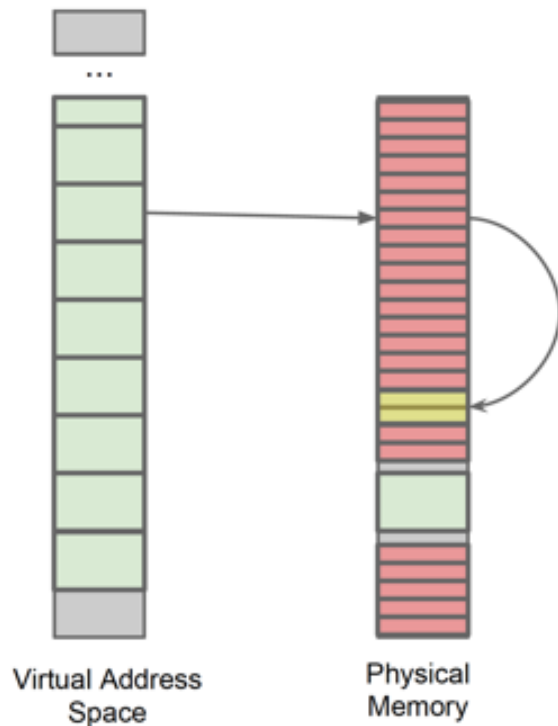
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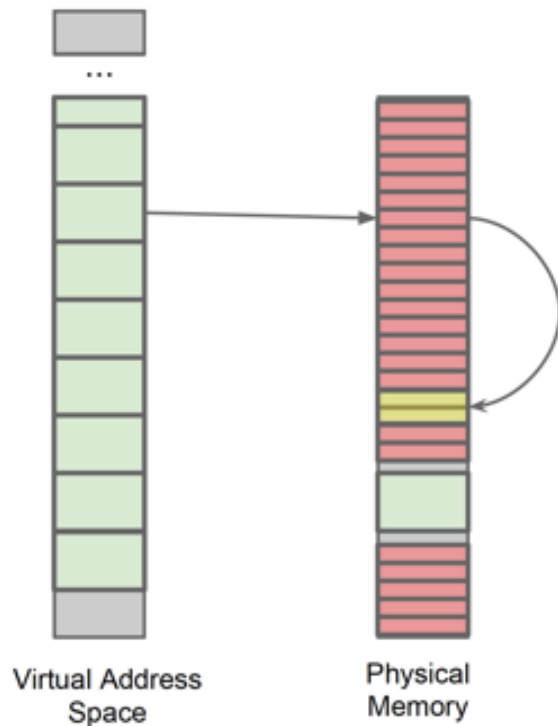
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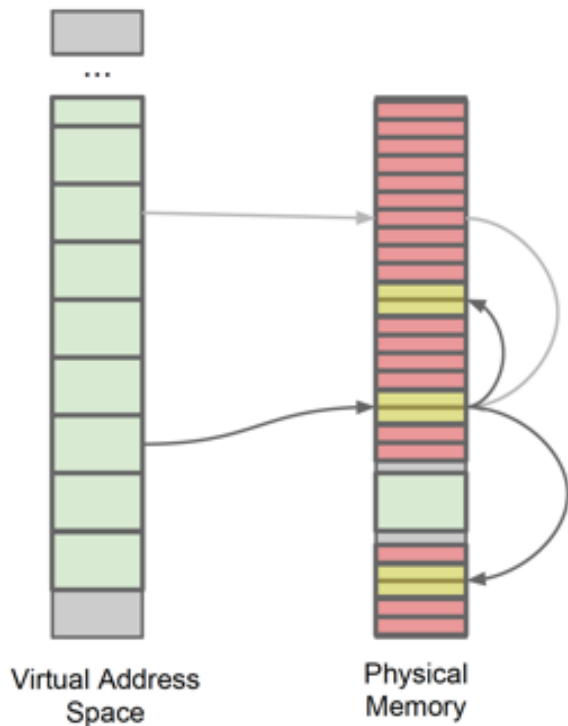
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An attacker that can read / write page tables can use that to map **any** memory read-write.

Exploit strategy

Privilege escalation in 7 easy steps ...

1. Allocate a large chunk of memory
2. Search for locations prone to flipping
3. Check if they fall into the “right spot” in a PTE for allowing the exploit
4. Return that particular area of memory to the operating system
5. Force OS to re-use the memory for PTEs by allocating massive quantities of address space
6. Cause the bitflip - shift PTE to point into page table
7. Abuse R/W access to all of physical memory

In practice, there are many complications.

More info

- ◇ Computer Architecture, Fall 2021, Lecture 5
 - ◇ RowHammer (ETH Zürich, Fall 2021)
 - ◇ <https://www.youtube.com/watch?v=7wVKnPj3NVw&list=PL5Q2soXY2Zi-Mnk1PxjEIG32HAGILkTOF&index=5>
- ◇ Computer Architecture, Fall 2021, Lecture 6
 - ◇ RowHammer and Secure & Reliable Memory (ETH Zürich, Fall 2021)
 - ◇ <https://www.youtube.com/watch?v=HNd4skQrt6I&list=PL5Q2soXY2Zi-Mnk1PxjEIG32HAGILkTOF&index=6>
- ◇ Rowhammer attack from Google
 - ◇ <https://www.blackhat.com/docs/us-15/materials/us-15-Seaborn-Exploiting-The-DRAM-Rowhammer-Bug-To-Gain-Kernel-Privileges.pdf>
 - ◇ <https://www.youtube.com/watch?v=0U7511Fb4to>

References

- ◇ Materials from
 - ◇ Onur Mutlu, ETH Zurich